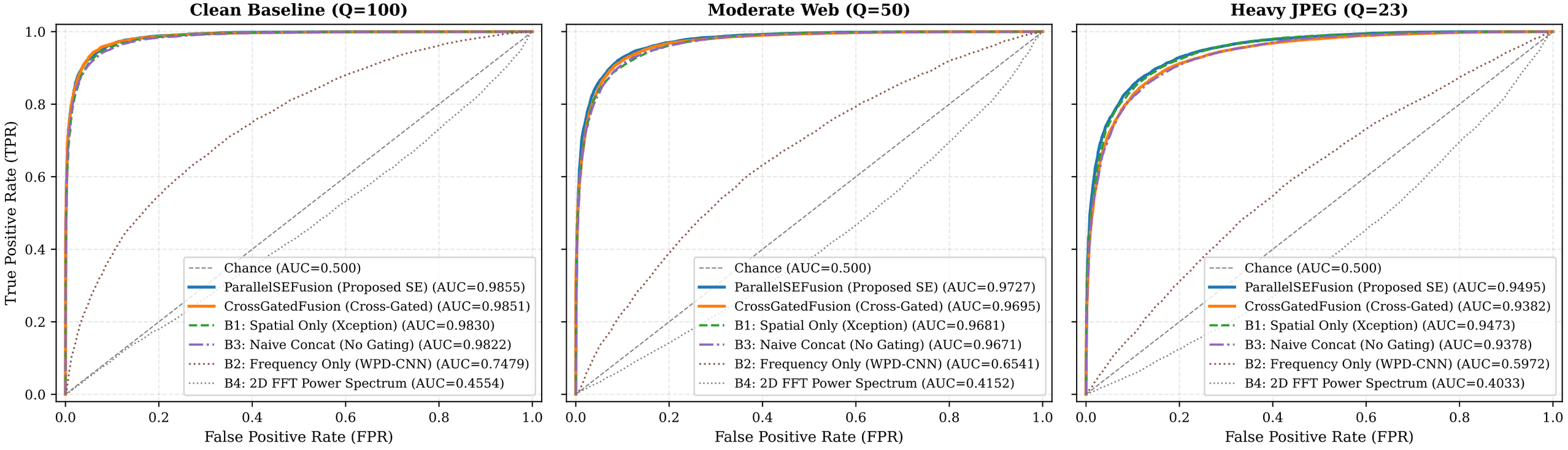

Controlled Receiver Operating Characteristic (ROC) Comparison across JPEG Quality Tiers
Clean Baseline (Q=100)
Moderate Web (Q=50)
Heavy JPEG (Q=23)
True Positive Rate (TPR)
False Positive Rate (FPR)
0.0
0.2
0.4
0.6
0.8
1.0
Chance (AUC=0.500)
ParallelSEFusion (Proposed SE) (AUC=0.9855)
CrossGatedFusion (Cross-Gated) (AUC=0.9851)
B1: Spatial Only (Xception) (AUC=0.9830)
B3: Naive Concat (No Gating) (AUC=0.9822)
B2: Frequency Only (WPD-CNN) (AUC=0.7479)
B4: 2D FFT Power Spectrum (AUC=0.4554)
Chance (AUC=0.500)
ParallelSEFusion (Proposed SE) (AUC=0.9727)
CrossGatedFusion (Cross-Gated) (AUC=0.9695)
B1: Spatial Only (Xception) (AUC=0.9681)
B3: Naive Concat (No Gating) (AUC=0.9671)
B2: Frequency Only (WPD-CNN) (AUC=0.6541)
B4: 2D FFT Power Spectrum (AUC=0.4152)
Chance (AUC=0.500)
ParallelSEFusion (Proposed SE) (AUC=0.9495)
CrossGatedFusion (Cross-Gated) (AUC=0.9382)
B1: Spatial Only (Xception) (AUC=0.9473)
B3: Naive Concat (No Gating) (AUC=0.9378)
B2: Frequency Only (WPD-CNN) (AUC=0.5972)
B4: 2D FFT Power Spectrum (AUC=0.4033)